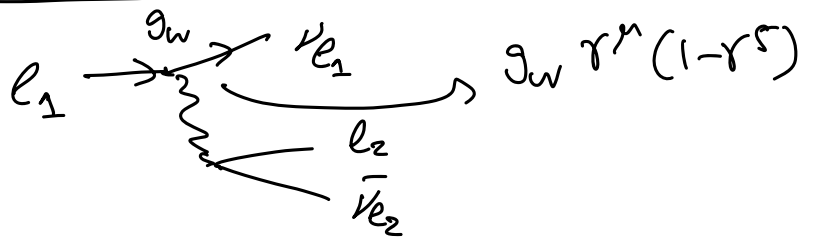


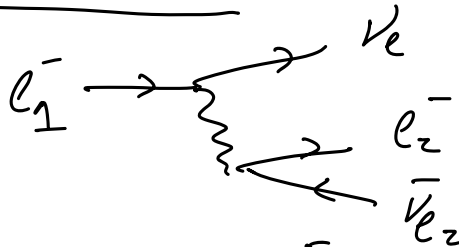
LFU in charged weak current

$$g_w^e = g_w^\mu = g_w^\tau = g_w$$



Prediction of τ lifetime

V-A theory



$$l_1 = \tau, \mu.$$

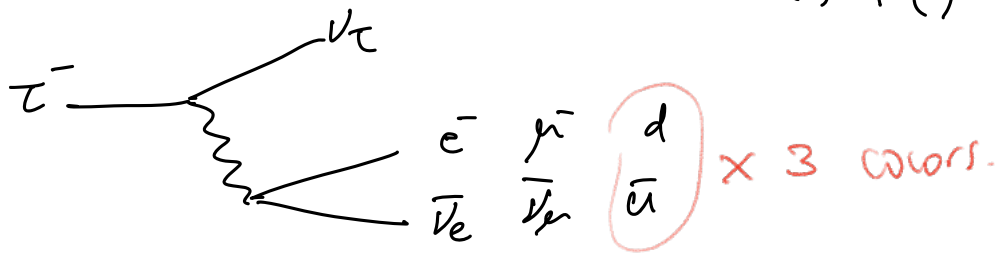
$$l_2 = e$$

$$\Gamma(l_1 \rightarrow e \bar{\nu}_e \nu_e) \propto G_F^2 m_e^5 \underbrace{\quad}_{\text{phase space}}$$

$$\frac{\Gamma(\tau \rightarrow e \bar{\nu}_e \nu_\tau)}{\Gamma(\mu \rightarrow e \bar{\nu}_e \nu_\mu)} = \frac{G_F^2}{G_F^2} \frac{m_\tau^5}{m_\mu^5} \frac{\rho(\tau \rightarrow e)}{\rho(\mu \rightarrow e)}$$

$$\mu \rightarrow e \bar{\nu}_e \nu_\mu \simeq 100\% \Rightarrow \Gamma(\mu \rightarrow e \bar{\nu}_e \nu_\mu) = \underbrace{\text{BF}(\mu \rightarrow e \bar{\nu}_e \nu_\mu)}_{\simeq 100\%} \Gamma_{\text{tot}}^\mu$$

$$\Rightarrow \Gamma(\mu \rightarrow e \bar{\nu}_e \nu_\mu) \simeq \Gamma_{\text{tot}}^\mu = \frac{1}{\tau_\mu}$$



$$\Gamma_{\text{tot}}^\tau \simeq \Gamma_e + \Gamma_\mu + \Gamma_{\bar{u}d} \times 3$$

$$m_\tau \gg m_e, m_\mu, m_u, m_d \Rightarrow \Gamma_{\tau \rightarrow e} \simeq \Gamma_{\tau \rightarrow \mu} \simeq \Gamma_{\tau \rightarrow \bar{u}d}$$

$$\Rightarrow \Gamma_{\text{tot}}^\tau \simeq 3 \Gamma_{\tau \rightarrow e} \Rightarrow \Gamma_{\tau \rightarrow e} \simeq \frac{1}{3} \Gamma_{\text{tot}}^\tau = \frac{1}{3} \frac{1}{\tau_\tau}$$

hypothesis/approximation.

$$\frac{\Gamma(\tau \rightarrow e \bar{\nu}_e \nu_\tau)}{\Gamma(\mu \rightarrow e \bar{\nu}_e \nu_\mu)} \simeq \frac{1}{3} \frac{\Gamma_{\text{tot}}^\tau}{\Gamma_{\text{tot}}^\mu} = \frac{1}{3} \frac{1}{\tau_\tau} \frac{\tau_\mu}{1} = \frac{m_\tau^5}{m_\mu^5} \times \frac{\rho(\tau \rightarrow e)}{\rho(\mu \rightarrow e)}$$

$$m_\tau \gg m_e, m_\mu$$

$$\simeq 1$$

$$\tau_{\tau}^{\text{pred}} = \frac{\tau_{\mu}}{5} \times \frac{m_{\mu}^5}{m_{\tau}^5}$$

$$\tau_{\mu} = 2.2 \times 10^{-6} \text{ s}$$

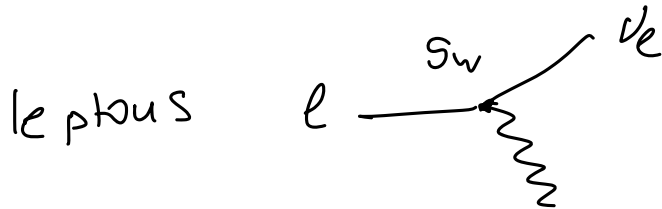
$$m_{\mu} = 106 \text{ MeV}$$

$$m_{\tau} = 1.78 \text{ GeV}$$

$$\Rightarrow \tau_{\tau}^{\text{pred}} = 3.3 \times 10^{-13} \text{ sec}$$

$$\tau_{\tau}^{\text{PDG}} = 3.1 \times 10^{-13} \text{ sec}$$

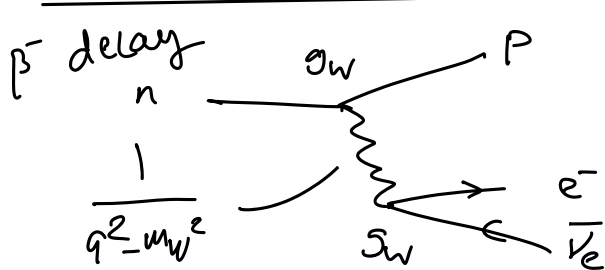
Good prediction of V-A with many approximation.



$$g_W \gamma^{\mu} (1 - \gamma^5)$$

Charged weak current

Weak interaction in hadrons



$$q^2 \approx (m_n - m_p)^2 \ll m_W^2$$

$$M_W \approx g_W^2 \bar{f}_p \gamma^{\mu} (1 - \gamma^5) f_n \frac{1}{q^2 - m_W^2} \bar{l}_e \gamma^{\mu} (1 - \gamma^5) l_e$$

initial state: $n \xRightarrow{=}$ $\xRightarrow{<=}$ 2 spin states

Final state: $p \xRightarrow{=}$ $\xRightarrow{<=}$

$e^- \xRightarrow{=}$ $\xRightarrow{<=}$

$\bar{l}_e \xRightarrow{=}$

$$P(n \rightarrow p e^- \bar{l}_e) \approx \Gamma_{\text{tot}}^n = \frac{1}{\tau_n} \Rightarrow \tau_n^{\text{pred}} \approx 1318 \text{ sec}$$

$$\text{Experimentally } \tau_n^{\text{meas}} = 878 \text{ sec} \neq \tau_n^{\text{pred.}}$$

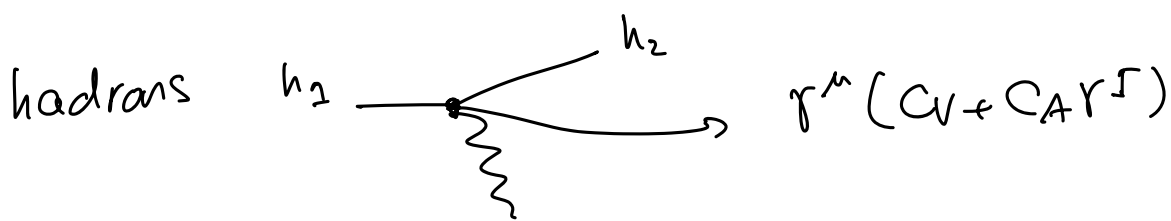
$\Rightarrow$ hadrons have a different behaviour

$$\Rightarrow \text{hadrons } \gamma^{\mu} (C_V + C_A \gamma^5)$$

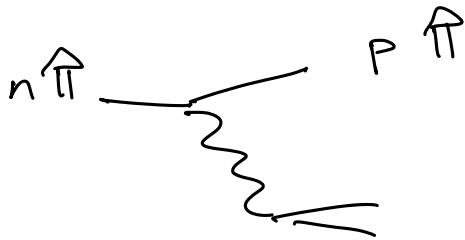
C_V : vector coupling

C_A : axial coupling

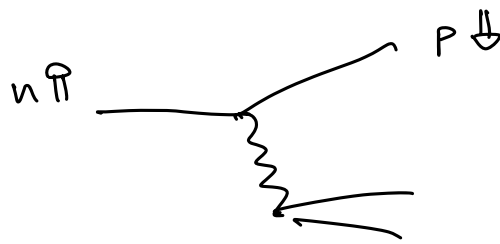
C_V, C_A two distinct coupling constants.



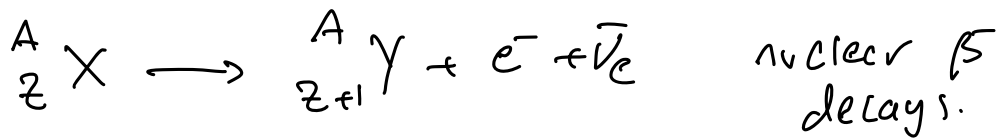
C_V, C_A : depend on hadrons considered



Fermi transition
dominated by C_V



Gamow-Teller transitions
sensitive to both C_V, C_A



$$\frac{d\Gamma(X \rightarrow \gamma e^- \bar{\nu}_e)}{dE_e} = \left(\frac{g_w}{m_w}\right)^4 \underbrace{f(C_V, C_A)}_{\text{comes matrix elements for different } X, Y}$$

Experimental picture:

$C_V \approx 1.000$ same as leptons

axial part depends on hadrons

neutron $C_A = -1.267$

similar to leptons
within 30%

neutron
 β decay

$$C_A/C_V = -1.267$$

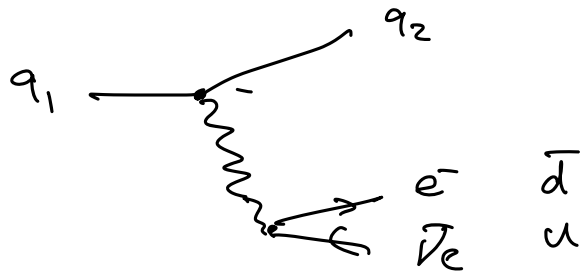
$$\Lambda \rightarrow p \pi^-, n \pi^0 \quad C_A/C_V = -0.718 \quad \Delta S = 1$$

$$\Lambda \equiv (uds)$$

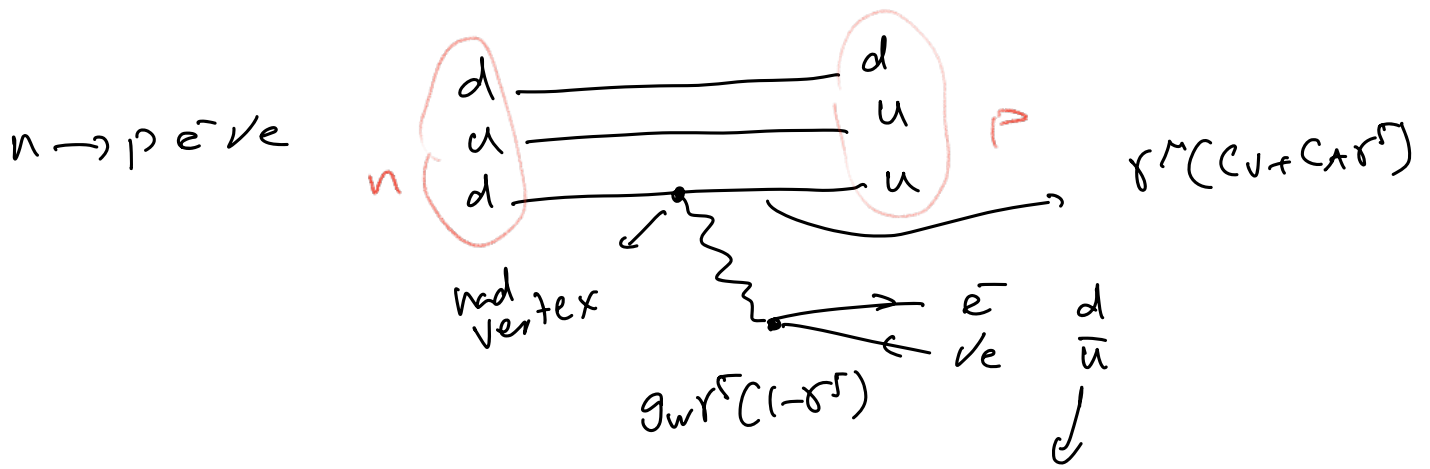
$$\Sigma^- \rightarrow n e^- \bar{\nu}_e \quad C_A/C_V = +0.340 \quad \Delta S = 1$$

$$\Xi^- \rightarrow \Lambda e^- \bar{\nu}_e \quad C_A/C_V = -0.250$$

$\Rightarrow$ Consider charge weak current for quarks



$$n \rightarrow p \pi^-$$

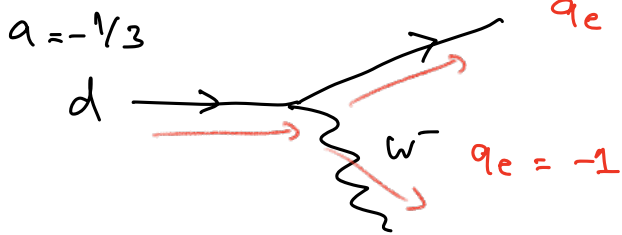


1 Simple leptonic vertex.

$$C_V = -C_A = 1$$

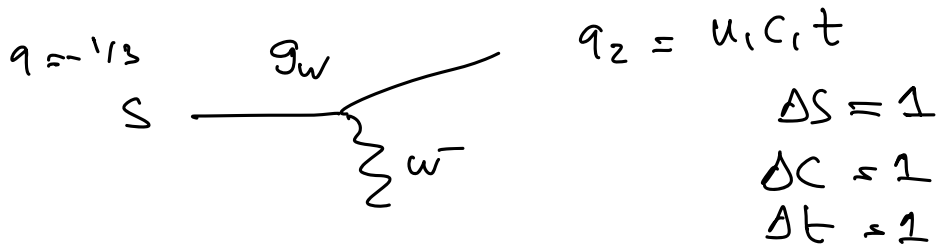
1 hadronic vertex that depends on C_V, C_A

Charge weak current with quarks :



$$q_e = +2/3 \Rightarrow q = u, c, t$$

$\Rightarrow$ No conservation of quark flavor



$$q_2 = u, c, t$$

$$\Delta S = 1$$

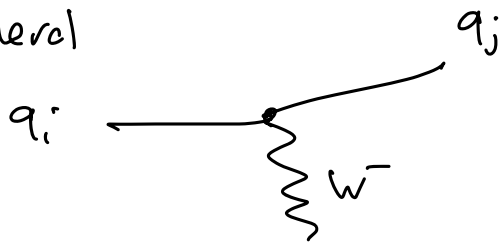
$$\Delta C = 1$$

$$\Delta b = 1$$

$$g_w^{su}, g_w^{sc}, g_w^{st}$$



In general



$g_w \underbrace{V_{ij}} \gamma^\mu (1 - \gamma^5)$
 V_{ij} : elements of
 Cab. bho- Kobayashi
 - Maskawa
 matrix

$$\begin{array}{c} u \\ c \\ t \end{array} \begin{array}{ccc} d & s & b \\ \left(\begin{array}{ccc} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{array} \right) \end{array}$$